2024 Calculus II Additional Exercise (12)

1. Exercise 33, in page 49 from Calculus Supplement Homework.

2. Exercise 8, 12 in page 53 from Calculus Supplement Homework.

3. Exercise 17 in page 54 from Calculus Supplement Homework.

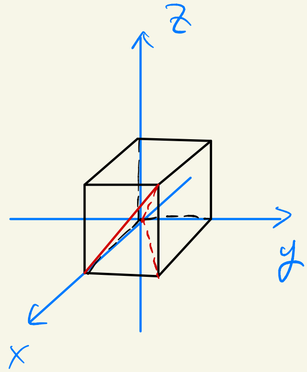
4. Exercise 28 in page 55 from Calculus Supplement Homework.

5. Exercise 32 in page 55 from Calculus Supplement Homework.

Q1 P49 Ex33

33. ★ Compute that

$$\int_0^1 \int_0^x \int_0^y \frac{\sin z}{(1-z)^2}\, dz dy dx.$$



解 由于被积函数没有显式原函数，可以通过交换积分次序来计算。首先交换y和z，这时将x看作常量。

$$\int_0^1 dx \int_0^x dy \int_0^y \frac{\sin z}{(1-z)^2} dz$$

$$= \int_0^1 dx \int_0^x dz \int_z^x \frac{\sin z}{(1-z)^2} dy$$

$$= \int_0^1 dx \int_0^x \frac{\sin z}{(1-z)^2} (x-z) dz$$

此积分被积函数还是不能积出，继续交换积分次序：

$$\int_0^1 dx \int_0^x \frac{\sin z}{(1-z)^2} (x-z) dz$$

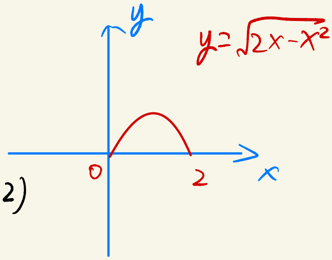
$$= \int_0^1 dz \int_z^1 \frac{\sin z}{(1-z)^2} (x-z) dx$$

$$= \int_0^1 \frac{\sin z}{(1-z)^2} \left[\frac{1}{2}(x-z)\right]_z^1 dz$$

$$= \frac{1}{2}\int_0^1 \sin z\, dz = \frac{1}{2}(1-\cos 1)$$

Q2 P53 Ex8

8. The centroid of the curve $y = \sqrt{2x - x^2}$ $(0 \le x \le 2)$ is (   ).



Sol: curve $y=\sqrt{2x-x^2}$

参数化 $\Longrightarrow \begin{cases} x = t \\ y = \sqrt{2t-t^2} \end{cases} \Rightarrow \begin{cases} x'(t) = 1 \\ y'(t) = \dfrac{1-t}{\sqrt{2t-t^2}} \end{cases}$ $(0 \le t \le 2)$

$$\Longrightarrow r'(t) = x'(t)\vec{i} + y'(t)\vec{j} = \vec{i} + \frac{1-t}{\sqrt{2t-t^2}}\vec{j}$$

$$\Longrightarrow \frac{ds}{dt} = |r'(t)| = \sqrt{1+\frac{(1-t)^2}{2t-t^2}} = \frac{1}{\sqrt{2t-t^2}}$$

$$\therefore M = \int_C ds = \int_0^2 \frac{1}{\sqrt{2t-t^2}}\,dt = \int_0^2 \frac{1}{\sqrt{1-(t-1)^2}}\,dt = \arcsin(t-1)\Big|_0^2 = \pi$$

$$M_y = \int_C x\,ds = \int_0^2 \frac{t}{\sqrt{2t-t^2}}\,dt = \int_0^2 \frac{t-1}{\sqrt{2t-t^2}} + \frac{1}{\sqrt{2t-t^2}}\,dt$$

$$= \int_0^2 \frac{d(t-1)^2}{\sqrt{1-(t-1)^2}} + \pi = -\sqrt{1-(t-1)^2}\Big|_0^2 + \pi = \pi$$

$$M_x = \int_C y\,ds = \int_0^2 1\,dt = 2$$

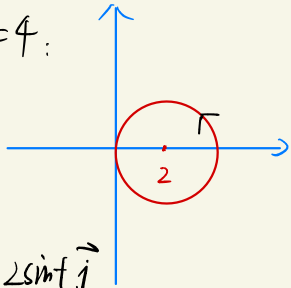
$\therefore$ centroid $\begin{cases} \bar{x} = \dfrac{M_y}{M} = 1 \\ \bar{y} = \dfrac{M_x}{M} = \dfrac{2}{\pi} \end{cases}$

□

P53 Ex12

12. Let L be the circle $x^2 + y^2 = 4x$, then $\oint_L \sqrt{x^2+y^2}\, ds = (\qquad)$.

Sol: parametrize closed curve L: $(x-2)^2+y^2=4$.



$$\begin{cases} \frac{x-2}{2} = \cos t \\ \frac{y}{2} = \sin t \end{cases} \Rightarrow \begin{cases} x = 2\cos t + 2 \\ y = 2\sin t \end{cases}$$

即曲线 $\gamma(t) = x(t)\vec{i} + y(t)\vec{j} = (2\cos t + 2)\vec{i} + 2\sin t\,\vec{j}$

$\gamma'(t) = -2\sin t\,\vec{i} + 2\cos t\,\vec{j}$

$|\gamma'(t)| \equiv 2$

$$\therefore \oint_L \sqrt{x^2+y^2}\, ds = \int_0^{2\pi} \sqrt{[x(t)]^2+[y(t)]^2}\cdot|\gamma'(t)|\,dt$$

$$= \int_0^{2\pi} \sqrt{8+8\cos t}\cdot 2\, dt$$

$$= \int_0^{2\pi} \sqrt{16\cos^2\frac{t}{2}}\cdot 2\, dt$$

$$= 8\left(\int_0^{\pi} \cos\frac{t}{2}\, dt - \int_{\pi}^{2\pi} \cos\frac{t}{2}\, dt\right)$$

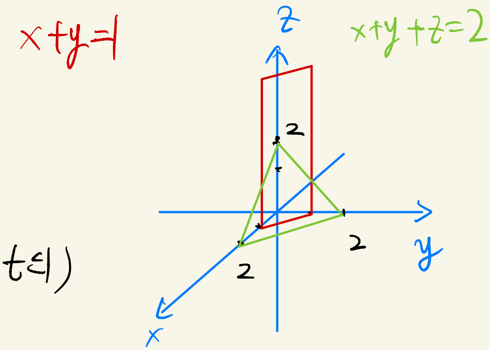
$$= 16\left(\sin\frac{t}{2}\Big|_0^{\pi} - \sin\frac{t}{2}\Big|_{\pi}^{2\pi}\right)$$

$$= 32$$

$\square$

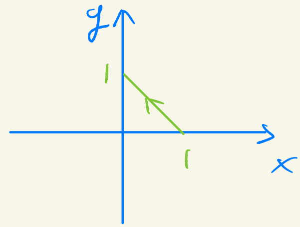
Q3 P54 Ex17

17. Compute $\oint_C \boldsymbol{F} \cdot d\boldsymbol{r}$, where $\boldsymbol{F}(x,y) = (y^2 - z^2, 2z^2 - x^2, 3x^2 - y^2)$ and C is the intersection of plane $x + y + z = 2$ and cylinder $|x| + |y| = 1$. In the counter-clockwise direction if seen from the positive direction of z.



Sol: 分xy的四个象限讨论，
先找相交直线方程再积分.

I: $\begin{cases} x+y+z=2 \\ x+y=1 \end{cases} \Rightarrow \begin{cases} x=1-t \\ y=t \\ z=1 \end{cases} \quad (0\le t\le 1)$



$\Rightarrow \vec{r}'(t) = -\vec{i} + \vec{j}$

$$\therefore \int_{C_1} \vec{F}\cdot d\vec{r} = \int_0^1 \vec{F}(x(t), y(t), z(t))\cdot \vec{r}'(t)\, dt$$

$$= \int_0^1 -\{[y(t)]^2 - [z(t)]^2\} + \{2[z(t)]^2 - [x(t)]^2\}\, dt$$

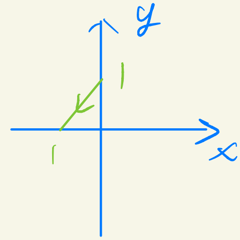
$$= \int_0^1 -\{(1-t)^2 - 1\} + \{2 - t^2\}\, dt$$

$$= -\frac{(t-1)^3}{3} - \frac{t^3}{3}\Big|_0^1 + 3$$

$$= (0 - \frac{1}{3}) - (-\frac{-1}{3} - 0) + 3$$

$$= \frac{7}{3}$$

II: $\begin{cases} x+y+z=2 \\ -x+y=1 \end{cases} \Rightarrow \begin{cases} x=-t \\ y=1-t \\ z=2t+1 \end{cases} \quad (0 \le t \le 1)$



$$\Rightarrow \vec{r}(t) = -\vec{i} - \vec{j} + 2\vec{k}$$

$$\therefore \int_{C_2} \vec{F}\, d\vec{r} = \int_0^1 -(y^2-z^2) - (2z^2-x^2) + 2(3x^2-y^2)\, dt$$

$$= \int_0^1 7x^2 - 3y^2 - z^2\, dt$$

$$= \int_0^1 7t^2 - 3(t-1)^2 - (2t+1)^2\, dt$$

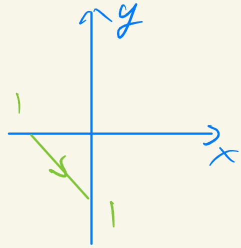
$$= \frac{7t^3}{3} - (t-1)^3 - \frac{(2t+1)^3}{6} \Big|_0^1$$

$$= \left(\frac{7}{3} - 0 - \frac{27}{6}\right) - \left(0 - (-1) - \frac{1}{6}\right)$$

$$= \frac{7}{3} - \frac{13}{3} - 1$$

$$= -3$$

III: $\begin{cases} x+y+z=2 \\ -x-y=1 \end{cases} \Rightarrow \begin{cases} x=t-1 \\ y=-t \\ z=3 \end{cases} \quad (0 \le t \le 1)$



$$\Rightarrow \vec{r}\,'(t) = \vec{i} - \vec{j}$$

$$\therefore \int_{C_3} \vec{F}\, d\vec{r} = \int_0^1 (y^2-z^2) - (2z^2-x^2)\, dt$$

$$= \int_0^1 x^2+y^2-3z^2\, dt$$

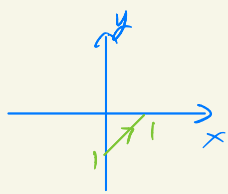
$$= \int_0^1 (t-1)^2+t^2-27\, dt$$

$$= \frac{(t-1)^3}{3} + \frac{t^3}{3}\Big|_0^1 - 27$$

$$= \left(0+\frac{1}{3}\right) - \left(-\frac{1}{3}+0\right) - 27$$

$$= \frac{2}{3} - 27$$

IV: $\begin{cases} x+y+z=2 \\ x-y=1 \end{cases} \Rightarrow \begin{cases} x=t \\ y=t-1 \\ z=3-2t \end{cases} \quad (0 \le t \le 1)$



$$\Rightarrow \vec{r}\,'(t) = \vec{i} + \vec{j} - 2\vec{k}$$

$$\therefore \int_{C_4} \vec{F}\, d\vec{r} = \int_0^1 y^2-z^2+2z^2-x^2-2(3x^2-y^2)\, dt$$

$$= \int_0^1 -7x^2+3y^2+z^2\, dt$$

$$= \int_0^1 -7t^2 + 3(t-1)^2 + (3-2t)^2 \, dt$$

$$= -\frac{7t^3}{3} + (t-1)^3 + \frac{(2t-3)^3}{6} \Big|_0^1$$

$$= \left(-\frac{7}{3} + 0 - \frac{1}{6}\right) - \left(0 - 1 - \frac{27}{6}\right)$$

$$= 1 + \frac{27}{6} - \frac{14}{6} - \frac{1}{6}$$

$$= 3$$

Finally, $\oint_C \vec{F} d\vec{r} = \left(\int_{C_1} + \int_{C_2} + \int_{C_3} + \int_{C_4}\right) \vec{F} d\vec{r}$

$$= \frac{7}{3} - 3 + \frac{2}{3} - 27 + 3$$

$$= -24$$

$\square$

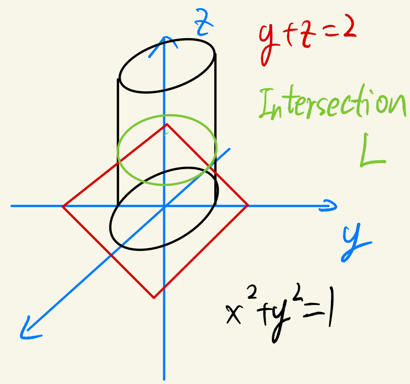
Q4 P55 Ex28

28. Let L be the intersection of $y + z = 2$ and $x^2 + y^2 = 1$ in the counterclockwise direction if view from the positive part of z-axis. Compute

$$\oint_L -y^2\,\mathrm{d}x + x\,\mathrm{d}y + z^2\,\mathrm{d}z.$$

Sol:

$$\begin{cases} x^2+y^2=1 \\ y+z=2 \end{cases} \xrightarrow{\text{prarmetrize}} \begin{cases} x=\cos t \\ y=\sin t \\ z=2-\sin t \end{cases} \quad (0\le t\le 2\pi)$$



$$\Rightarrow \begin{cases} x'(t) = -\sin t \\ y'(t) = \cos t \\ z'(t) = -\cos t \end{cases}$$

$$\therefore \oint_L -y^2dx + xdy + z^2dz$$

$$= \int_0^{2\pi} -y^2x' + xy' + z^2z'\,dt$$

$$= \int_0^{2\pi} \sin^3 t + \cos^2 t - (2-\sin t)^2\cos t\,dt$$

$$= \int_0^{2\pi} \cos^2 t - 1\,d\cos t + \int_0^{2\pi} \frac{\cos 2t + 1}{2}\,dt - \int_0^{2\pi} (\sin t - 2)^2\,d\sin t$$

$$= \frac{\cos^3 t}{3} - \cos t\Big|_0^{2\pi} + \pi - \frac{(\sin t - 2)^3}{3}\Big|_0^{2\pi}$$

$$= \pi$$

Q5. P55 Ex32

32. Let C be the segment on the curve $x = 1 - y^2$ from $(0, -1)$ to $(0, 1)$, compute

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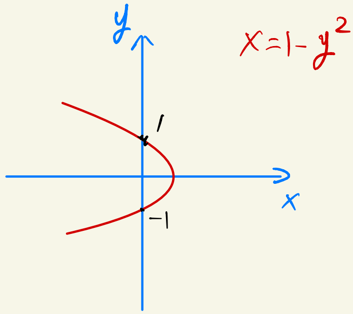
$$\int_C y^2\,\mathrm{d}x + x^2\,\mathrm{d}y.$$



Sol: $x=1-y^2 \xrightarrow{\text{参数化}} \begin{cases} x=1-t^2 \\ y=t \end{cases} \quad (-1\le t\le 1)$

$\Rightarrow \begin{cases} x'(t)=-2t \\ y'(t)=1 \end{cases}$

$$\therefore \int_C y^2dx+x^2dy = \int_{-1}^{1} y^2x'+x^2y'\,dt$$

$$= \int_{-1}^{1} -2t^3+(1-t^2)^2\,dt$$

$$= \int_{-1}^{1} t^4-2t^3-2t^2+1\,dt$$

$$= \frac{t^5}{5}-\frac{2t^3}{3}+t\Big|_{-1}^{1}$$

$$= \frac{2}{5}-\frac{4}{3}+2$$

$$= \frac{16}{15}$$

$\square$